

ARL Theory Question answers

Name: John Maged Naiem

ID:2501293

1. Working principles of LiDAR, camera and radar

LiDAR:

- it sends laser light rays into FoV and then these rays reflect from objects and return back to the LiDAR sensor and measure the time difference between emitting and receiving rays
- It calculates the distance of the object from it by using this equation [Distance = (Speed of light x Time elapsed) / 2]
- It produces a point cloud of the objects in 3D space and used to define geometry of this objects
- It used to measure distances, angles, coordinates of the object and some types is used to find colors and texture of objects using the reflected intensities patterns but not one of the main data returns of LiDAR
- Algorithms like Euclidean clustering can be used for clustering and it requires high computation level as millions of light rays/ second are be processed

Radar:

- Sends radio waves into a FoV and receives reflected rays from objects
- Uses similar methods in LiDAR to measure distances and uses doppler effect to measure the velocity of the moving object as reflected frequency changes compared to the emitted one
- Measures angles, distances and velocities of objects
- Depend of RCS of material as the high RCS material like metals is more visible in radars than lower ones like plastics

Camera (Passive):

- It receives only light rays from the ones reflected on the objects and these rays passes through lenses capturing them in sensors in tiny pixels
- It records visual info about the objects like shape and colors
- The pixels measure the amount of light falls on each pixel creating the image and have info about RGB (red, green, blue)
- It only capture coordinates of objects in the pixel not in real world that why it doesn't measure distance or anything, only a 2D collection of pixels of the image

- Algorithms and AI like SIFT and YOLOv2 can be used to help extract meaning info about the object in the image for classification
-

2. 3D positions by 2 cameras

- The cameras catch the image at same time and the pixels of the left image is found to be identical on the right in the right image but shifted a little, this shift is called disparity
 - using this disparity as well as the distance between both cameras lens and focal length the depth is calculated for this pixel using triangulation geometry
 - The calculation is repeated for all the pixels and a 3D point cloud is formed for the 2D image captured on camera
 - Software algorithms like SIFT is used to find the identical features in each image and other algorithms like KNN is used to match them from left image and right image
 - Rectification ensures that the identical pixels is on the same horizontal line in both images
-

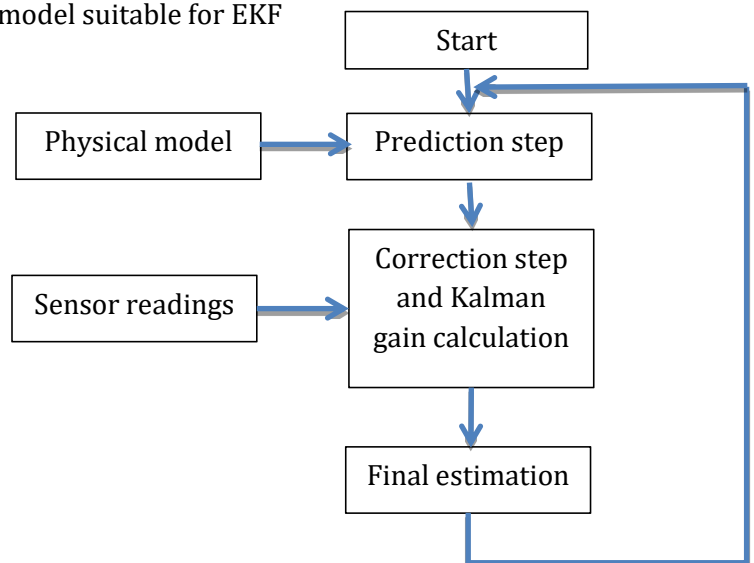
3. Machine Learning

- Neural networks are machine learning models which uses layers of mathematical units called neurons and consists of 3 parts input which data are entered , hidden layers where processing happen and output which gives out the final decision
- It learn by recognizing patterns and features from past train data and uses weights and biases for processing
- Learning process happen in 4 steps :
 - 1)Forward propagation: model takes input data, process it through hidden layers and an output prediction is done
 - 2)error calculation: a loss function is used to calculate the difference between the prediction and true data
 - 3)backpropagation: the model tracks back and calculates how much each weight and bias of each neuron contributes in this error
 - 4)optimization: the model fine tune each weight and bias so it is as close as possible to the true value using optimization methods like gradient decent
- This cycle is repeated every time new training data enter the model
- This model can be used in identification and classification of objects in images when using cameras as well as used in keypoint algorithm to know depth of the objects in mono camera

- Can be used in LiDAR in detection and classification of the point clouds into 3D objects
- Can be used to collect data from many sensors (sensor fusion) like LiDAR, radars, cameras, IMU to give a more accurate prediction of object info like its distance, velocity and its color so better perception

4. Kalman Filter

- It is a estimation algorithm which uses the last physical model and sensor measurements to to accurately estimate as both the model and sensors have noise and uncertainly so we cant fully trust only one source
- It has 2 steps : 1)prediction based on the last model
2) correction and update after receiving new sensor data
- a value called Kalman Gain weight each the prediction and sensor measurement and decide which to trust use more in their final estimation
- for the car in unknown environment I would use EKF as normal KF is only for linear models but this car would have non linear model suitable for EKF
- I would estimate: 1)Velocity
2)X and Y position
3) heading direction
4) yaw rate
5) acceleration
- Sensors to be used will be: 1)LiDAR
2)camera
3) GNSS
4)IMU
5)GSS



5. SLAM

- SLAM is an algorithm that is used to map and localize the object it is connected to in an unknown map at the same time and some types are EKF SLAM, , FastSLAM 2.0, and graph SLAM.
- SLAM uses the sensors like LiDAR and cameras to detect the landmarks making the map and their locations and uses the motion related sensors like IMU and GSS and

GNSS to know the car location compared to the initial position especially between the sensor readings

- When the SLAM finds landmarks already seen it optimizes their position in the map and optimizes the pose of the vehicle compared to this landmark
- FastSLAM 2.0: it uses a particle filter method that use sensor readings to detect the landmarks and the physical model to create many particles then weight each particle according to the new sensor readings and highest weighted particles are duplicated and focused on
- EKF SLAM: it uses the physical model to predict the pose of the vehicle then sensor readings are received about landmarks observations which is used to update the estimation of the pose and the landmarks positions

6. Graph Search Algorithms

BFS	DFS	Dijkstra's	A*
It starts by visiting all the neighbouring from root node in first level then do the same for each neighbouring node visiting the nodes in next level of this node visiting them level by level	It starts by visiting all the nodes in a branch till finding a dead end or goal is found and then track back this branch and explore other branches emerging from this visited branch so it visits them branch by branch	It is used for weighted graphs and done by 2 step cycle exploring and updating cost of nodes, exploring is done for nodes which have the lost cost to reach them from root node and updating is checking if there is a lower cost for neighbouring nodes than the last calculated one until the lowest cost path is found for goal	It is very similar to Dijkstra's but uses heuristic with the cost as well and the finds the next node to visit which have the lowest sum of cost and heuristic and heuristic is the estimated distance to the goal.
Ensures that shortest path to the goal is found	Uses low memory	Good with weighted graphs and ensures the shortest path on them	Faster than Dijkstra's and ensures lowest path
Takes a lot of	doesn't ensure that	Requires more	Requires the

memory and takes time in large maps	this is the shortest path to goal	processing and time in large maps	coordinates of the goal before hand and used for single goal only and high memory usage
-------------------------------------	-----------------------------------	-----------------------------------	---

7. Practical Motion Planning

- Using path smoothing as a post processing so that after the A* maps find the shortest required path to the goal it checks between 2 nodes of the zigzag path drawn can be skipped with a line joining both nodes directly
- That increases the risk of collision with wall and obstacles as corner cutting the zigzag path will make the robot risks hitting the obstacles as it might not account for the robot size and shape

8. Control Theory and Philosophy

- Control means doing a set of commands to change the output of a dynamic system to a desired output
- Control theory is a mathematical branch which focus on making systems which controls a dynamic system making them have a steady output as needed in lowest time possible with no error in the process and used in aerospace for example maintaining the steady state of a plane in air and in cruise control on cars maintaining speed and in automatic insulin pumps which helps in maintaining insulin level in blood compared to sugar amount
- Open loop system: it doesnt depend on the output of the system but works on a condition stated beforehand like time and doesn't check if goal is reached or not just does the process ex: microwave heating food for time disregarding the temperature of the food
- Closed loop system: it does depend on the output and compares it with the desired output and corrects itself to minimize the difference and does this by feedback ex: automatic head light in cars
- Feedback is the process of returning the output of the system back as an input for the controller so it can compare the difference error and correct it so the desired output is reached

- An air conditioner unit records the temperature of the room and returns it back to the controller to compare with the desired temperature and minimizes the difference error to achieve this temperature

9. PID control

- PID controller consists of 3 elements: 1)proportional
2)integral
3)derivative
- This controller is a closed loop system which compares the actual process variable produced by system and compares it with the setpoint set beforehand and calculates the error between them and this loop is repeated while using feedback
- The error calculated is reduced in 3 ways:
- Proportional: it produces an output directly proportional to the error magnitude larger error larger output
- Integral: it sum all the past errors to remove the steady state error remaining
- Derivative: it predict the the error in the future based on the rate of change in the error before and slows down the system when near the set point avoiding overshoot
- Tuning methods: manual tuning , Ziegler-Nichols method
- Manual tuning: it is a trail and error method which first the P increased until oscillations are seen then the I is increased to slowly remove the steady state error then the D is increased to slow the oscillations and prevent overshoot
- Ziegler-Nichols method: the I and D is set to zero the the P is increased till constant amplitude oscillations are seen(ultimate gain K_u) and then measure the oscillations time period and use the Ziegler-Nichols' standard formulas to calculate the P , I, D from K_u and Time period
- Temperature furnace system: I would set the set point to the required temperature then use the sensor readings to input it to the PID controller to calculate the difference error between the recorded temperature and set point one and then change the power in the heating element to reach the set point
- Limitations: it is a linear control system and the D is sensitive to noise and hands only one input and one output cant work with multiple processing variables
- MPC control system can be used with non linear and complex scenarios as well and in scenarios where multiple variables are

to tracked and in systems have high time delay PID will not be suitable

10. References

- AMZ driverless paper
- Sciencedirect.com
- va-imaging.com
- GeeksforGeeks.com
- IBM.com
- mathworks.com
- https://youtu.be/8m0VP5_feOY?si=WbWlMGBg2IlR5j8v
- https://youtu.be/EFg3u_E6eHU?si=gmKq8Qjr94SpovNc
- https://youtu.be/zaBhtODEL0w?si=DpFq9juY_0cuSL8N
- <https://youtu.be/mwn8xhgNpFY?si=gcck13T9-MwbPBLf>

